



AQELYN PLATFORM

Cyber Security Operating Environment (CSOE)

Building the world's most understandable cyber intelligence environment

Architecture Baseline v1.0 | Approved for Implementation

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At a Glance

Product	AQELYN Platform
Category	Cyber Security Operating Environment (CSOE)
Core approach	AI-native, evidence-first, explainable, human-centered
Target users	Private users, SMBs, enterprises, and government organizations
Primary coverage	Endpoints, identity, cloud, web, networks, attack surface, vulnerabilities, evidence, compliance, and remediation
Deployment modes	Local-only, managed cloud, enterprise SOC, private cloud, hybrid, and air-gapped
Engineering baseline	Engineering Archive packages EA-0001 through EA-0064
Core principle	Every finding must be understandable by a non-expert and actionable by an expert

Executive Summary

AQELYN is an AI-native Cyber Security Operating Environment that helps individuals, businesses, enterprises, and government organizations understand, secure, and continuously improve their digital environments. It combines evidence-based intelligence, explainable AI, and safe, approval-based remediation in a single platform.

Rather than functioning as an antivirus product, vulnerability scanner, dashboard, SIEM, XDR, SOC console, or point solution, the platform operates as a unified environment that continuously discovers and analyzes digital assets across endpoints, identities, networks, cloud services, websites, applications, containers, and supported IoT and edge devices.

Traditional security tools generate alerts and technical reports, then leave interpretation to the reader. AQELYN takes the opposite approach: understanding comes before action. Each finding explains what happened, why it matters, how the conclusion was reached, how serious the issue is, what to do about it, and whether the platform can safely carry out that action on the user's behalf.

Internally, information is correlated through a centralized Knowledge Graph, validated by an Evidence Engine, evaluated against trust and policy models, and delivered through an Explainability Engine. The result is security intelligence that non-experts can read and act on, without sacrificing the technical depth, evidence, and auditability that professionals require. A single modular platform scales from private endpoint assessment and browser security analysis through cloud posture management, identity protection, attack surface management, vulnerability assessment, and compliance monitoring, up to workflow automation, risk analysis, reporting, and AI-assisted remediation under explicit approval.

Why AQELYN Exists

Most organizations run a patchwork of disconnected security products. The result is fragmented data, duplicate alerts, operational overhead, and persistent uncertainty about what to fix first.

AQELYN replaces that fragmentation with a single intelligent operating environment. It discovers assets and exposures continuously, correlates signals across identity, endpoint, cloud, network, and web, and explains its findings in clear language backed by verifiable evidence. Problems are prioritized by risk and context, remediation is guided or safely automated under policy, and each fix is verified after execution. Over time this closes the loop between detection and measurable improvement.

The objective is not simply to detect threats. It is to help people and organizations understand and resolve security problems with confidence.

Vision, Mission, and Product Promise

Vision. Become the world's most trusted Cyber Security Operating Environment by making cybersecurity understandable, explainable, evidence-based, and actionable for everyone.

Mission. Empower every user, from individuals to governments, to understand, protect, and continuously improve their digital environment through AI-driven, evidence-based cyber intelligence.

Motto. Understand. Protect. Improve.

Product promise. AQELYN finds security problems, explains them clearly, proves them with evidence, and guides or safely automates remediation for private users, businesses, enterprises, and government organizations.

Product Principles

Explain before you recommend. A finding is not complete until it states what happened, why it matters, how the conclusion was determined, which evidence supports it, the likely impact of inaction, the recommended actions, and the expected outcome once the issue is resolved.

Simplicity first. Security information is presented in language appropriate to the reader's knowledge level. Complex terminology is explained, never assumed.

Evidence before opinion. Findings and recommendations are backed by collected evidence, policy evaluation, threat intelligence, risk calculation, or reproducible engineering logic.

Human-centered security. The platform reduces uncertainty, avoids alarmist messaging, and helps users make confident decisions.

Expert depth on demand. Any simplified finding can be progressively expanded into technical detail, raw evidence, policy evaluation, event history, and engineering context.

These principles reduce to a single rule: **every finding must be understandable by a non-expert and actionable by an expert.** They are reinforced by platform-wide commitments to privacy by design, zero trust by default, transparency, auditability, least privilege, secure defaults, data minimization, and human oversight for high-impact automation.

What Makes AQELYN Different

One unified operating environment. Capabilities that are normally spread across separate products — endpoint security intelligence, identity and governance, cloud and infrastructure security, web intelligence, attack surface management, vulnerability management, evidence collection, risk analysis,

compliance and policy, workflow automation, safe remediation, and reporting — operate on a shared data model within a single environment.

Explainable problem finding. Each finding answers five questions: What is wrong? Why is it risky? How serious is it? How do I fix it? Can AQELYN fix it for me?

Human-readable remediation. Findings surface as understandable risk cards with practical actions: Fix Now, Show Me Why, Ignore with Reason, Schedule Fix, and Create Ticket.

Evidence-first architecture. Findings link to evidence, recommendations explain their reasoning, and every remediation action is traceable and auditable.

Adaptive operating modes. The same platform supports local-only private use, centrally managed business deployments, and large-scale enterprise and government operations.

Architecture-driven development. Development follows a structured Engineering Archive baseline, which provides requirement traceability, consistent architecture, controlled implementation, and long-term maintainability and scalability.

Security Domains

The platform is organized around four integrated security domains.

1. Endpoint Security

Protection and assessment cover the full range of endpoints found in modern environments:

- Windows PCs and laptops
- Linux desktops, laptops, and servers
- macOS systems
- Android devices, plus iPhone and iPad where platform restrictions allow
- Tablets, servers, and virtual machines
- Cloud workloads, containers, and Kubernetes environments
- Supported IoT and edge devices

2. Identity and Governance

This domain manages identities, authentication, authorization, and privileged access, and evaluates identity risk for both human and machine identities. It also covers policy management, governance, compliance, access review, and trust relationships.

3. Cloud and Infrastructure Security

Coverage extends to public and private cloud workloads, virtual machines, containers and Kubernetes, networks and storage, infrastructure as code, cloud identities and permissions, and configuration posture across multi-cloud and hybrid environments.

4. Threat Intelligence and Attack Surface Management

Continuous attack surface discovery and external exposure monitoring provide visibility into internet-facing assets. Threat intelligence and vulnerability correlation add context, so risk is prioritized against the environment it actually affects. Change detection and remediation tracking keep that picture

current as the environment evolves.

Core Capabilities

On the discovery and analysis side, the platform provides continuous asset discovery and inventory, endpoint security assessment, web intelligence, and posture management for identity, cloud, and the external attack surface, together with vulnerability discovery and correlation.

Collected information is preserved with full provenance, correlated in the Knowledge Graph, and evaluated against trust and policy models to produce risk scoring, prioritization, and compliance and governance monitoring.

On the action side, explainable AI recommendations feed approval-based remediation, workflow and case management, reporting and dashboards, and complete audit trails. An API, SDK, and plugin layer supports integration with existing tooling.

Endpoint Security Intelligence

The Endpoint Security Assessment Engine performs intelligent analysis directly on supported devices, using an agent or another appropriate collection mechanism where the operating system permits it.

Assessment covers the software and execution environment, including running processes, services, installed software, startup entries and autoruns, and device drivers. Local security posture is evaluated through certificates, accounts and privileges, security policies, firewall configuration, and the state of Microsoft Defender or third-party antivirus products. Browser extensions and stored password metadata are analyzed without collecting secrets.

The engine also inspects USB and removable devices, network configuration, and open ports, along with virtualization and container platforms such as Docker, Kubernetes, Windows Subsystem for Linux, and Hyper-V. Operating system and security event logs round out the picture.

Web Intelligence Engine

The Web Intelligence Engine performs authorized security analysis of external and web-facing assets.

Transport and infrastructure checks include HTTP header analysis, TLS configuration assessment, certificate-chain validation, DNS analysis, WHOIS and registration metadata, subdomain discovery, and authorized open-port analysis. At the application layer, the engine evaluates Content Security Policy, HTTP Strict Transport Security, and the SPF, DKIM, and DMARC email authentication records, inspects robots.txt and sitemap.xml, and follows redirect chains.

Technology fingerprinting and CMS identification feed CVE correlation and the detection of exposed administrative interfaces, while monitoring of externally visible configuration changes keeps results current.

All scanning is authorized, policy-governed, rate-limited, auditable, and designed to avoid disruption to the systems under assessment.

Safe AI and Remediation

Fully autonomous remediation creates operational and security risk when actions are taken without transparency, guardrails, or appropriate approval. The platform therefore uses an approval-based automation model: automated actions must be explainable, evidence-based, governed by policy, auditable, reversible where practical, approved according to user or organizational rules, and validated after execution.

Automation is introduced gradually. Low-risk, reversible actions may be automated under policy, while high-impact actions require explicit approval or elevated governance.

Operating and Deployment Modes

Private mode is designed for individual users. Operation is local-only where practical, with no mandatory cloud connectivity, privacy-first defaults, user-controlled telemetry, and options for local evidence storage and analysis.

Business mode adds managed cloud services, centralized administration, multi-device management, shared policy and reporting, and integration with ticketing and workflow systems.

Enterprise and government mode supports multi-tenant or dedicated deployment, SOC and SIEM integration, governance, risk, and compliance, fine-grained access control, data residency controls, private cloud and hybrid deployment, air-gapped installation where required, and enterprise-scale automation and audit.

Supported deployment models range from stand-alone desktop and home network installations, through small business and public, private, and hybrid cloud deployments, to dedicated government environments and air-gapped sites.

How AQELYN Works

The platform operates as a continuous pipeline:

Discover → Collect → Normalize → Correlate → Analyze → Explain → Recommend → Approve → Remediate → Verify → Improve

Discovery identifies devices, identities, services, websites, cloud resources, and exposures. Authorized telemetry, configuration, events, and evidence are collected, normalized into the Universal Object Model, and correlated in the Knowledge Graph alongside identities and risk. Analysis then applies policy, trust, threat intelligence, vulnerability, and behavioral models.

Results are explained — what was found, why it matters, and how the conclusion was reached — and translated into prioritized, practical remediation options. Policy and human approval gates govern execution. Approved actions are carried out or guided step by step, then verified to confirm that the action succeeded and risk decreased. Outcomes feed back into the platform, continuously strengthening security posture over time.

High-Level Architecture

A modular, event-driven, evidence-first architecture underpins the platform. At its core, the AQELYN Kernel coordinates the Universal Object Model, the Event Bus, the Evidence Engine, and the Knowledge Graph. Around this core sit the Trust, Mission, Workflow, and Policy engines, identity and governance services, the Endpoint and Web Intelligence engines, and the cloud and attack surface services.

AI orchestration and explanation services generate recommendations, API, SDK, plugin, and integration layers expose the platform to external systems, and observability, audit, and reporting services provide operational insight.

The design targets secure-by-default operation, least privilege, modular deployment, horizontal scaling, and controlled extensibility.

Complete Digital Visibility

Contextual visibility extends across identities and users, endpoints, workstations, and laptops, servers and virtual machines, containers and Kubernetes, cloud workloads and services, networks and storage, websites and web applications, internet-facing assets, browsers and extensions, and supported IoT and edge devices — together with the findings, evidence, policies, and remediation history associated with each of them.

Engineering Approach

Development follows a documented enterprise architecture consisting of Engineering Archive packages EA-0001 through EA-0064. The archive covers requirements, architecture, security, AI, APIs, databases, governance, deployment, coding standards, design systems, developer workflows, engineering portal design, and final readiness controls.

Each implementation maps to an approved architecture package and its associated requirements. Changes are version-controlled, reviewed, tested, and traceable. This architecture-first approach enables AI-assisted software engineering while preserving consistency, maintainability, and long-term governance.

Implementation and AI-Assisted Development

Implementation follows a fixed workflow. Developers and AI agents begin with README.md and START_HERE.md, then read the Project Charter and engineering standards, and review EA-0058 through EA-0061 as global development rules. EA packages are implemented sequentially, beginning with EA-0001, with Codex as the primary implementation agent and Claude Code as an independent architecture and code reviewer.

Automated tests, security checks, and architecture compliance checks run on every change, and human approval is required before any production merge. Traceability, documentation, and implementation status are updated with each approved change.

No EA package may be skipped, and the repository structure may not be redesigned without an approved architecture change.

Repository Structure

The repository uses the following approved, fixed hierarchy:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md  
■■■■ START_HERE.md
```

README.md provides the public, project-level overview. START_HERE.md is the mandatory implementation entry point for developers and AI coding agents. The archive/ directory contains the authoritative Engineering Archive packages.

Quick Start

For Readers and Stakeholders

Start with this README and review the executive vision, product principles, and platform domains. The Architecture Atlas provides visual orientation, and the relevant Engineering Archive packages contain the detailed specifications.

For Developers and AI Coding Agents

1. Open START_HERE.md and read the Project Charter, engineering principles, repository standard, architecture guide, and development rules.
2. Read EA-0058, EA-0059, EA-0060, and EA-0061.
3. Create an implementation branch for EA-0001 and implement only the approved scope.
4. Add unit, integration, security, and acceptance tests.
5. Update traceability and documentation.
6. Submit the change for Claude Code, automated, and human review.
7. Merge only after all gates pass, then continue sequentially to the next EA package.

Definition of Done

A capability is complete only when the approved requirements are implemented, the code compiles and passes static checks, and the unit, integration, and security tests pass. Performance requirements must be met, evidence and audit behavior verified, and user-facing explanations confirmed against the product principles. Documentation and traceability must be current, architecture compliance confirmed, and review and approval complete.

Roadmap

Milestone	Status
Architecture and planning baseline	Complete
Product identity and design direction	Complete
Implementation foundation	Next
Core runtime engines	Planned
Endpoint, identity, cloud, web, and attack surface capabilities	Planned
Integrated alpha	Planned
Controlled beta	Planned
Production v1.0	Planned
Continuous improvement and regulated-market expansion	Future

Security, Privacy, and Responsible Use

Systems may only be scanned or assessed with explicit authorization from the user or organization that owns them.

The platform applies least privilege, explicit consent, data minimization, encryption in transit and at rest, clear retention controls, user-controlled telemetry, audit logging, and secure secrets handling. Local, privacy-preserving operation is preferred where practical, and policy and approval gates govern all high-impact actions.

The platform exists to improve defensive security and digital trust. It must not be used for unauthorized access, disruption, surveillance, or other harmful activity.

Contributing

Contribution rules are defined in the AQELYN Developer Handbook and Coding Standards. Contributions must reference the applicable EA package and requirements, preserve the fixed repository structure, and include tests and documentation. They must follow secure coding requirements, pass the CI/CD and architecture compliance gates, avoid introducing undocumented functionality, and receive the required review and approval before merge.

Documentation

Key entry points:

README.md	Product and project overview
START_HERE.md	Implementation entry point
archive/	Engineering Archive source of truth
docs/	Supporting documentation
api/	API specifications and implementation
sdk/	Developer SDKs
plugins/	Approved extensions
examples/	Reference implementations and usage examples

License and Legal Status

The final software license, trademark policy, privacy policy, terms of service, and third-party notices will be added before public distribution or production release.

AQELYN is a proposed product and brand identity. Trademark clearance and legal registration should be completed in the relevant jurisdictions before commercial launch.

Closing Statement

More than a security product, AQELYN is a Cyber Security Operating Environment built to make digital security understandable, evidence-based, explainable, and actionable for everyone.

Understand. Protect. Improve.